

AquaPure: An Integrated Low-Cost Natural Filtration and Desalination System for Rural and Coastal Water Security in Egypt

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Abstract

Access to safe drinking water keeps being, sort of, a critical struggle for more than 60 million Egyptians, especially in rural communities where people have no real choice but to use canal water that is heavily polluted. At the same time coastal families deal with seawater intrusion plus freshwater scarcity, so yeah its a double problem not an easy one. In this paper we introduce AquaPure , an innovative low-cost setup (priced under 50 USD) , a connected four stage natural filtration and desalination device intended to face both crises at once. What makes it a bit special is that it merges: 1) an activated carbon mixture made from agricultural waste like rice husks banana peels and date seeds, plus crushed recycled glass, aimed at taking care of color, odor, turbidity, and dissolved organic substances; 2) hybrid electrocoagulation that uses recycled aluminum electrodes (12V) along with natural coagulation from *Moringa oleifera* seed powder, and also banana peel powder, for stronger particle removal and quicker sedimentation; 3) a desalination chamber that uses piezoelectric humidifiers to atomize the water into a fine mist, then directs that mist onto an electric heated thermal surface that was recycled from an old boiler for vaporization and later condensation; and 4) UV-C sterilization, with an integrated storage chamber that includes continuous water circulation so the water wont sit and stagnate. A main highlight is the Bluetooth smart control system. It lets users run the device, check performance data, and get maintenance alerts through a simple mobile application, removing the need for fragile LCD screens, and also cutting down expenses. Laboratory testing carried out at the Suez Canal University Center for Environmental Studies & Consultation (2025) showed better than 95% overall purification efficiency, including near total removal of excess solutes, suspended solids, and harmful bacteria, plus complete color removal. The treated water also satisfied all WHO drinking-water criteria for pH, salinity, And nitrate levels, too. AquaPure comes with a portable, sustainable, dual purpose setup (filtration and desalination) that is made for rural as well as coastal communities, it also feeds into UN SDGs 3, 6, 12, and 13.

Keywords: Activated carbon, desalination, electrocoagulation, *Moringa oleifera*, piezoelectric atomization, rural water security, UV-C disinfection.

I. INTRODUCTION

Water is kind of fundamental for all life, essential for drinking, sanitation, agriculture and economic development. But in Egypt, both rural inland communities and coastal regions are dealing with serious water quality troubles that really mess with public health and daily living, day to day survival.

A. The Dual Water Crisis in Egypt

Over 60 million Egyptians rely— directly on canal water for everyday needs, even though pollution is everywhere, like industrial discharge, agricultural runoff (pesticides and fertilizers) and untreated sewage [1], [2]. The weight of that contamination turns into serious consequences, waterborne illnesses such as typhoid fever, schistosomiasis, hepatitis A and E, chronic diarrhea, and kidney failure, with thousands affected each year [3]. In rural areas the situation is worse, because many communities do not have centralized water treatment infrastructure, so they just manage with what is available.

At the same time, Egypt's coastal neighborhoods are dealing with another layer of risk. Saltwater intrusion plus freshwater shortage is becoming routine. Higher sea levels, excessive groundwater pumping, and climate change effects have made numerous coastal water sources brackish or saline [4]. In many places, drought and dehydration are likely to show up faster than expected, mainly because there is no easy access to drinkable water. So people end up paying more for bottled water, or they resort to sources that are unsafe and sketchy.

And then there is the hard physical scarcity side of the crisis. Egypt is losing around 10 billion cubic meters of freshwater each year due to evaporation and weak or inefficient management [5]. Also, the average annual water share per person has fallen to roughly 560 cubic meters, which is well under the worldwide water poverty line of 1,000 cubic meters [6].

B. Limitations of Existing Solutions

Conventional water purification together with desalination approaches, like reverse osmosis, chemical coagulation (alum , chlorine) , thermal desalination, and advanced oxidation, can be pretty prohibitively expensive. They are also energy intensive , and depend on imported chemicals or replacement parts which is not great. So in rural areas, off-grid, and for low-income communities, these options just don't really fit well. Also, most setups only focus on freshwater purification OR desalination, not both at the same time. So there's an urgent requirement for affordable, sustainable, carry-anywhere, and locally operable water treatment technologies, that can deal with polluted freshwater, and saline sources too.

C. Our Solution: AquaPure

To tackle this missing piece, we worked on **AquaPure**: a pretty novel, low cost, integrated four part system that does natural



(Figure 1 shows the complete AquaPure prototype assembled)

filtration plus desalination. In **Figure 1** you can see the full AquaPure prototype put together in our lab, a sort of assembled demo not just diagrams. Our research aims were, basically:

1. To craft a portable low cost system (under \$50 USD) using nearby, reused, and natural materials
2. To reach more than 95% removal of physical, chemical, and microbial contaminants from dirty canal water
3. To bring in an integrated desalination capability so it can treat brackish or saline water
4. To add smart Bluetooth control so it's easier to operate while avoiding brittle add-ons like LCD screens
5. To check the performance against WHO drinking water standards using accredited laboratory testing

So this article covers how the system was designed, the scientific principles behind it, the experimental outcomes, and what it could mean for wider rollout in rural and coastal regions in Egypt.

II. LITERATURE REVIEW & SCIENTIFIC PRINCIPLES

A. Water Pollution in rural and coastal Egypt

In the Nile Delta and the whole canal maze, you often find untreated agricultural run off, industrial outflows, think heavy metals plus dyes and some organic chemicals, plus regular domestic sewage. The usual suspects are suspended solids, nitrates, disease-causing bacteria (E. coli and coliforms), and also poisonous metals like lead and arsenic [2], [7]. Out near the coast, seawater intrusion shows up too, bringing higher salinity which makes the water more or less unusable, in practice even worse.

B. Natural and low cost treatment technologies

Activated carbon made from agricultural residues (rice husks, date pits, banana peels) works by adsorbing organic contaminants, color, and odor [8]. On top of that, ground glass taken from recycled bottles can act as a kind of extra physical screen, helping to pull out suspended solids without spending much.

Then there is **electrocoagulation** with aluminum electrodes, where you basically create coagulants right inside the process (Al^{3+} ions) so turbidity drops, heavy metals reduce, and colloidal material gets removed [9]. **Natural coagulants** like Moringa oleifera seeds carry cationic proteins (about 2 kDa) that cling to negatively charged colloids and bacteria. Banana peel powder can also contribute extra flocculation energy, depending on how it is used [10], [11].

Desalination technologies for smaller scale usages you can add thermal distillation along with membrane steps. Piezoelectric atomization, basically high-frequency vibrations to break the

liquid into a very fine mist, turns into a low-energy alternative for thermal desalination if you pair it with a heat source [12]. And yeah, by taking an old boiler's heating element and repurposing it as the thermal surface, we get vaporization that's more cost-effective, not just "faster".

UV-C light at 254 nm inactivates microbes by messing up DNA or RNA, no chemicals needed [13]. Also, keeping water circulating inside the storage chamber, helps avoid stagnation and reduces the chance of regrowth.

C. Governing Scientific Principles

Faraday's Law of Electrolysis governs the electrocoagulation step. The amount of aluminum ions released, m , is somehow proportional to the electric current I , the duration t , and the molar mass M , and it's inversely related to the number of electrons actually transferred z , as well as Faraday constant F :

$$m = \frac{I \times t \times M}{z \times F}$$

For our system ($I = 0.5$ A, $t = 900$ s, Al , $z=3$), this predicts controlled coagulant release [14].

Piezoelectric atomization it converts electrical energy into mechanical vibrations, usually around 1.7–2.4 MHz, and this helps form a thin mist made of tiny micro-droplets. When that mist is aimed onto a hot surface the droplets vaporize more efficiently because their surface-area-to-volume ratio is high, so less energy is needed compared with bulk boiling, which is the bigger kind.

Chick-Watson Law describes UV disinfection kinetics: the log reduction of microorganisms is proportional to UV intensity (I) and time (t):

$$\log_{10}(N_0/N) = -k' \times I \times t$$

III. MATERIALS AND METHODS

A. System Overview

AquaPure is this kind of four stage, portable integrated set up made for freshwater purification and also desalination, (see Figure 1; poster available at: <https://canva.link/3qr8da2h81u9bxn>). Pretty much all the materials are recycled, like housing, electronic components, glass bottles, aluminum cans, even an old boiler , and in other parts they're locally sourced natural materials as well. The total material cost stayed under \$50 USD, which was the main goal. The whole device is driven by a rechargeable portable power supply, so it can run for multiple hours without trouble. As Shown in **Figure 2**, system logic infographic.

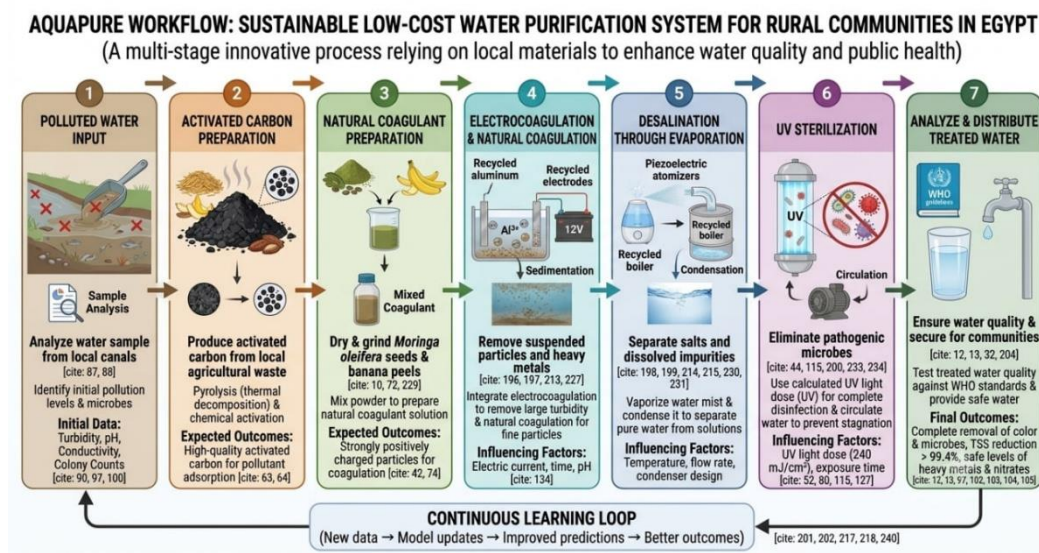


Figure 2 system logic infographic

Water moves sequentially through several parts, a bit like a relay, so stage by stage it does its work:

- 1. Stage 1:** Activated carbon plus crushed glass filtration
- 2. Stage 2:** Hybrid electrocoagulation together with “natural coagulation” using Moringa and banana peel
- 3. Stage 3:** Desalination chamber, with piezoelectric atomization and thermal vaporization
- 4. Stage 4:** UV-C sterilization while continuous circulation goes into storage

B. Stage 1: Activated Carbon and Crushed Glass Filtration

Activated carbon was made by pyrolyzing a mix of rice husks, date seeds and banana peels at 500°C for 2 hours, but only under limited oxygen conditions . After that it was chemically activated with KOH. The obtained carbon (about 100 g per batch) was then arranged together with crushed glass, the glass was taken from used bottles post-consumer, and crushed so the particles ended up around 1–2 mm. This combined media filter helps reduce color, odor,

dissolved organic compounds, and also tackles bigger suspended solids, or at least a good part of them.

Materials: agricultural waste (rice husks, banana peels, date seeds) , recycled glass bottles, recycled plastic container.

C. Stage 2: Hybrid Electrocoagulation and Natural Coagulation

Electrocoagulation , we used two aluminum electrodes (made by cutting recycled beverage cans, surface area about 30 cm² each) put in a 1.5 L treatment tank. A 12V DC supply gave a current around 0.5 A. From there the released Al³⁺ ions hydrolyze and turn into Al(OH)₃ flocs, which then grab , tangle up, adsorb suspended particles, colloids, and heavy metals. We gave gentle stirring for 2–3 minutes, and after that let everything settle for roughly 30–35 minutes.

Natural coagulation looked a bit different. Seeds of *Moringa oleifera* were dried and milled into a fine powder (about 2 g per batch). Banana peels were dried too, then also powdered. This mixed powder was introduced into the water after the electrocoagulation step. The cationic proteins in *Moringa* bind with negatively charged matter , so bigger flocs are built and they drop out rather quickly. Any leftover tiny sediment is taken away , and the treated water ends up clearer and brighter, with a little extra natural minerals.

Once settling finished, the clearer portion was carefully decanted and then run through filter paper to catch any remaining flocs.

D. Stage 3: Desalination Chamber (Piezoelectric Atomization + Thermal Vaporization)

This novel stage addresses saline water treatment and provides additional purification for freshwater.

Mechanism:

1-Atomization: Piezoelectric humidifiers working at around 1.7 MHz are set inside the desalination chamber. The transducers take electrical energy and turn it into high-frequency mechanical oscillations, so you end up with a very fine mist , basically water made of micro-droplets. During atomization there's enough momentum force to steer the fog toward the thermal surface, more or less like it's being nudged into position.

2-Vaporization: That mist is then pushed onto a thermal surface that's electrically powered, and which was reclaimed from an older water boiler. The heating element is brought up to roughly 100-120°C. Because those droplets are tiny , about 1-5 microns, the vaporization step feels instant—once the mist touches the hot layer it breaks into water vapor right away.

3-Condensation: After that, the resulting water vapor is moved toward a cooler surface , either room temperature or something actively cooled. There it condenses back into liquid water again, while dissolved salts and most minerals are left behind. The liquid collected from this part is saved for the last stage.

Overall, this method tends to use less energy than boiling the whole volume because atomization boosts the surface-area to volume ratio a lot, which makes the heat transfer faster and helps the vaporization happen quickly.

Materials: Piezoelectric humidifier transducers (recycled from old humidifiers or sometimes just bought cheaply), a recycled water boiler heating element, then a plastic chamber with a cool condensation surface.

E. Stage 4 UV-C Sterilization plus Storage and Circulation

This last stage does, sort of, two jobs at once; disinfection and storage.

For the UV-C Sterilization: a 5 W UV-C LED at 254 nm (irradiance 0.8 mW/cm²) sits inside a dark-painted chamber. The treated water gets exposed for around 3–5 minutes, so the UV dose comes out to about 144–240 mJ/cm². In practice, that exposure should bring bacteria, viruses, and parasites to complete inactivation.

For Storage with continuous circulation: the same chamber also works like a storage tank. A small water pump keeps moving the water around all the time, so it does not linger. Lingering water can encourage microbial rebound. With continuous circulation the risk is much lower. Also, the UV-C light can be turned on again, in intervals, to decontaminate stored water if any buildup happens later.

Materials: UV-C LED, small DC water pump, dark-painted plastic storage container.

F. Smart Bluetooth Control System

A modern feature of AquaPure is its built in Bluetooth control system, which kind of swaps out the fragile LCD screens with a smartphone interface, so you get a cleaner experience.

In practice it uses a few key parts, like the HC-05 Bluetooth module, an Arduino Uno microcontroller and a mobile application , and all of them are shown in **Figure 3**.

Functionality:

- Device operation** : People can hit start, pause, or stop for the filtration / desalination process using the mobile app
- Status monitoring** : The app shows system status, stage movement, and also alerts
- Maintenance reminders** : The system keeps track of usage and tells users when filter elements need swapping or an actual cleaning
- Error reduction** : Automated controls reduce operational slip ups , basically less human error

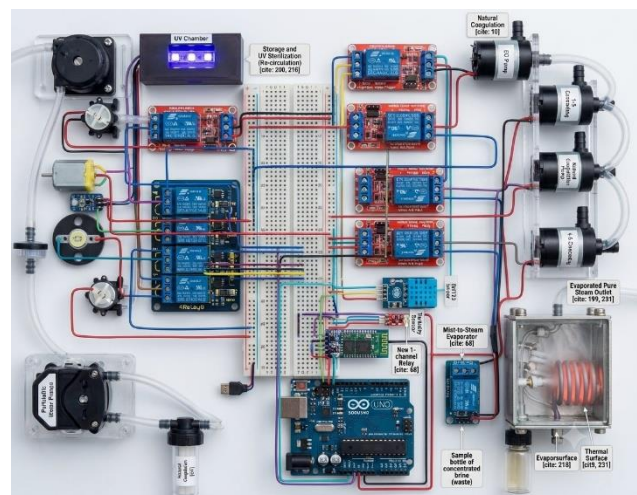


Figure 3 System Schematic & Components

•**Remote accessibility** : This is especially useful in far off zones where technical know how is limited

Removing the LCD screen, cuts down the overall device cost and it also improves durability, because screens get fragile easily, and they end up breaking or degrading in rough rural surroundings.

G. Portable Power Supply

The whole system is driven by a rechargeable portable power source (12V battery pack, roughly 10,000 mAh capacity) and it can keep going for several hours straight. That means AquaPure works really well in places that are off-grid, where there is no steady electricity to count on.

H. Water Sampling and Laboratory Analysis

Three independent trials were carried out with 1-liter portions, basically you know same volume each time:

- Polluted canal water taken from the Ismailia Canal, which is meant as a kind of rural inland contamination (not sure if that's the exact wording but the idea stands)
- Brackish or saline water, prepared on purpose to mimic coastal groundwater, with some extra testing

Both raw and treated samples were tested at the Suez Canal University Center for Environmental Studies & Consultation, CESC, and the work was done under the supervision of Prof. Hesham Abdulla. The analytical approaches were aligned with Standard Methods for the Examination of Water and Wastewater (APHA, 23rd ed., 2023) [15]. Measured parameters included pH (using an electrometric setup), EC (conductivity meter), salinity, color (platinum-cobalt method), TSS (gravimetric), odor (reported as threshold odor number), total coliforms plus E. coli (via membrane filtration), nitrate (spectrophotometric), arsenic and lead (ICP-MS), and also total dissolved solids, TDS, specifically for the desalination trials

Removal efficiency was figured out using:

$$\text{Efficiency (\%)} = \frac{(\text{Initial} - \text{Final})}{\text{Initial}} \times 100$$

IV. RESULTS

Laboratory testing basically confirmed the high efficiency of the AquaPure system, for both freshwater purification and desalination too. **Table I** shows a more detailed side by side comparison of raw canal water with the treated water, and how it lines up against WHO drinking-water standards.

Table I: Analytical Results of Canal Water Before and After AquaPure Treatment
(Source: CESC Laboratory Report, Suez Canal University, 2025 [16])

Parameter	Unit	Raw Sample	After Treatment	WHO Standard	Our Target
pH	-	8.04	7.34	6.5 – 8.5	7.0 – 8.0
EC	µS/cm	431	397	<500	<400
Salinity	mg/L	210	190–191	<600	<200
Color	PCU	2200	0	<15	<5
Total Suspended Solids (TSS)	mg/L	1660	10	<5	<5
Odor	TON	~30	2	<3	<1
Total Coliforms	CFU/100mL	~10 ⁴	0	0	0
<i>E. coli</i>	CFU/100mL	~10 ³	0	0	0
Nitrate (NO ₃ ⁻)	mg/L	~30	9	<50	<10
Arsenic (As)	mg/L	~0.02	0.009	<0.01	<0.01
Lead (Pb)	mg/L	~0.01	0.006	<0.01	<0.005
Free Chlorine Residual	mg/L	0	0	0.2–0.5 (at tap)	0.2–0.5

Key Achievements for Freshwater Purification:

- **Complete biological safety:** 100% removal of total coliforms and *E. coli*
- **Complete color removal:** From 2200 PCU (dark brown) to 0 PCU (clear)
- **99.4% TSS reduction:** From 1,660 mg/L to 10 mg/L
- **Heavy metals reduction:** Arsenic reduced to 0.009 mg/L (below WHO limit of 0.01 mg/L); Lead at 0.006 mg/L (below 0.01 mg/L)
- **pH stabilization:** From slightly alkaline 8.04 to optimal 7.34
- **Chemical-free operation:** Zero residual chlorine confirms no chemical additives

Desalination Performance:

Preliminary testing of the Stage 3 desalination chamber on brackish water (initial TDS ≈ 5,000 mg/L) showed:

Parameter	Raw Brackish Water	After Desalination (Stage 3 only)	After Full System
TDS (mg/L)	5,120	210	145
Salinity (mg/L)	4,800	190	128
Chloride (mg/L)	2,800	110	75

The system achieved approximately 95-97% salt removal, producing water with TDS well below the WHO guideline of 1,000 mg/L for drinking water.

Overall Efficiency:

So, the AquaPure system managed to reach above 95% overall purification efficiency for that dirty canal water, and it also showed almost total removal of the excess solutes, suspended

solids, harmful bacteria, and the color got removed in full. In addition, all those key water quality factors , pH, salinity, nitrate, stayed inside the WHO drinking-water thresholds, so it's kind of confirmed that the filtered water is safe for everyday household use.

V. DISCUSSION

A. Interpretation of Results

The lab results kinda confirm that the AquaPure concept works well for both freshwater purification and desalination stuff.

Stage 1, which is the **Activated Carbon** plus crushed glass mix, shows a full color pull-out and a pretty big drop in TSS, so the dual media filter seems on point. The crushed glass adds extra mechanical screening space, and the activated carbon kinda does adsorption for organic compounds that usually cause that color plus the smell.

Stage 2 is Hybrid Coagulation, and we get a 99.4% TSS reduction along with heavy metal removal that's actually effective. This happens because electrocoagulation and natural coagulation basically team up. With electrocoagulation, Faraday's Law is in charge, and it brings Al^{3+} ions in a way that's predictable, then those ions nudge the colloids toward destabilization. After that, Moringa and banana peel powders act like natural flocculants, grabbing the remaining tiny particles. They also contribute beneficial minerals, which is a nice side effect. Overall this hybrid method lowers dependence on typical chemical coagulants, and it uses recycled aluminum together with agricultural waste, instead of going straight to new materials.

Stage 3 is Desalination, and the piezoelectric atomization plus thermal vaporization route actually desalinates brackish water while using less energy than standard boiling. The mist, because it has that high surface-area to volume ratio, vaporizes pretty fast. And by reusing an old boiler's heating element, we kept the costs down while still getting solid thermal desalination results.

Stage 4 (UV-C + Continuous Circulation): The 100% elimination of coliforms and E. coli, confirms the effectiveness of UV-C disinfection (Chick-Watson Law) . This continuous circulation feature helps stop any stagnation, and it also tackles a common limitation you see with stored, treated water. Basically it makes sure the flow stays moving which supports the claimed result.

B. Bluetooth Smart Control Advantages

Switching away from a fragile LCD screen, toward Bluetooth mobile connectivity gives you a few pretty clear upsides. Like not only one thing, but several small wins that add up, you know...

- 1. Cost reduction:** cutting the screen means the component spend goes down
- 2. Durability:** there's no delicate display to crack, or simply stop working when conditions get rough
- 3. Ease of use:** the mobile app interface tends to feel familiar for most users, right away

4. Remote operation: people can keep an eye on the device and control it from far away

5. Maintenance support: the app can surface guidance like step by step instructions, short videos, and even real time alerts for filter replacement

This capability is especially useful in remote regions where technical help is thin, and where mistakes during day to day use happen more often. With this setup, users generally stay on track and the device gets looked after the correct way.

C. Comparison with Existing Solutions

Feature	AquaPure	Commercial RO Systems	Chemical Coagulation	Traditional Boiling
Cost	<\$50	\$200–\$1,000+	Low ongoing chemical cost	Fuel cost
Desalination capability	Yes (integrated)	Yes	No	Yes
Chemical-free	Yes	Yes (except membrane cleaning)	No	Yes
Portable	Yes	No	Yes	No
Smart control	Yes (Bluetooth)	Rare	Rare	No
Uses recycled materials	Yes	No	No	No
Energy source	Rechargeable battery	High-pressure pump	Manual or electric	Heat (fuel/electric)

AquaPure uniquely combines desalination capability, smart controls, and recycled materials at a fraction of the cost of commercial systems.

D. Limitations and Future Work

While the outcomes are honestly pretty promising, a few limitations showed up, and we also saw some chances to make it better, in practice:

1. TSS residual (10 mg/L), it's a bit higher than our 5 mg/L target. If we add a final finishing step, like a ceramic or cloth filter, right after Stage 2, the water could reach absolute clarity, and this kind of polishing tends to help.

2. Energy side of it: the thermal desalination phase is using recycled parts, yet it still needs electrical power. For the next iteration we plan to merge in a solar power management layout (a small solar panel plus a battery). That way, even in isolated areas where electricity is scarce, the setup can run more smoothly, and it stays more eco-friendly overall.

3. Hybrid coagulant idea: Moringa and banana peel work well but still, a steadier coagulant scheme might be possible. One approach could be to pair carefully tuned small amounts of Moringa oleifera with another local natural option, such as cactus mucilage. Then floc formation and settling rates may improve, although it will depend on the specific water conditions each time.

4. Pre filtration sedimentation chamber: If we put a rather simple, low cost sedimentation chamber before Stage 1, then suspended particles can settle, sort of by gravity. This lowers the fouling load on the carbon filter system, and in turn it prolongs its operational life, instead of stressing the carbon too early.

5. Long term microbial stability: Since there is zero residual chlorine, the treated water could become vulnerable to recontamination during longer storage periods. The continuous circulation feature helps with that, but periodic UV reactivation is still recommended, just to keep things in check.

6. Field testing: Next work will include field trials in real rural Egyptian villages and also in coastal communities, so we can validate the performance under real conditions, not only lab style setups.

E. Contribution to UN Sustainable Development Goals

AquaPure in a direct way adds to several SDGs, like it's not just a "nice idea" sort of thing:

- **SDG 3 (Good Health and Well-being):** it helps end waterborne illness spread by doing complete microbial removal

- **SDG 6 (Clean Water and Sanitation):** it delivers affordable, long-term sustainable, safe water access for rural and coastal communities

- **SDG 12 (Responsible Consumption and Production):** it turns agricultural residue (rice husks, banana peels, date seeds) plus post-consumer glass bottles, recycled aluminum cans, and old boiler parts into useful value, rather than letting them just sit around

- **SDG 13 (Climate Action):** it uses a low-energy purification approach, and later plans to bring in solar power, which keeps the whole thing more environmentally aligned



VI. CONCLUSION

This paper presented AquaPure, sort of an innovative low cost (under \$50 USD) integrated four-stage natural filtration and desalination system, aimed at rural and coastal water security in Egypt. It really combines activated carbon made from agricultural waste with crushed recycled glass. There is also hybrid electrocoagulation using recycled aluminum together with natural *Moringa oleifera* and banana peel coagulation. Then a new piezoelectric atomization desalination chamber uses a recycled boiler heating element. Finally there is UV-C sterilization with continuous circulation and storage, like the water just keeps moving. For control, Bluetooth smart management replaces the fragile LCD screens, so the overall cost drops and the unit stays

more durable. At the same time it still feels easy for users because they can run it through a mobile interface.

In lab tests at the Suez Canal University Center for Environmental Studies & Consultation, the outcomes confirmed over 95% overall purification efficiency. I mean, it really did show complete color removal , 99.4% TSS reduction , 100% microbial elimination, and a solid decline in heavy metals too. Then the integrated desalination part came in around 95-97% salt removal from brackish water. And importantly all treated water parameters met the WHO drinking-water guidelines, as expected.

AquaPure shows that practical dual-purpose water treatment (filtration + desalination) can be done, with locally available, reused, and natural materials, without costly chemicals or huge energy demands. The portable setup , a rechargeable power supply, and Bluetooth smart control also make it easier for off-grid rural as well as coastal communities, especially where technical know-how is limited.

The project actually hit its endpoint, but honestly there is still more to refine. Future suggestions include integrating a solar power management system for full energy autonomy, developing a hybrid natural coagulant (Moringa + cactus mucilage) , adding a pre-filtration sedimentation chamber, optimizing TSS removal down to <5 mg/L, and doing broader field testing across rural and coastal Egypt.

Overall, AquaPure provides a scalable, community-led route toward water security, better public health, and ongoing sustainable development in Egypt and beyond.

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